## Exercise C - II

### SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Find the coordinates of the points of intersection of the curves $y = \cos x$, $y = \sin 3x$ if $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$.

वक्रों $y = \cos x$ व $y = \sin 3x$ के प्रतिच्छेदन से बने बिन्दुओं के निर्देशांक ज्ञात करे यदि $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ .

**Ans.** $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right), \left(\frac{\pi}{8}, \cos\frac{\pi}{8}\right), \left(-\frac{3\pi}{8}, \cos\frac{3\pi}{8}\right)$

**Sol.** $y = \cos x; y = \sin 3x$

$\Rightarrow$ P.O.I. are

$\cos x = \sin 3x$

$\Rightarrow \cos\left(\frac{\pi}{2} - 3x\right) = \cos x$

$\Rightarrow \frac{\pi}{2} - 3x = 2n\pi \pm x; n \in I$

$\Rightarrow \frac{\pi}{2} - 3x = 2n\pi + x; n \in I$ or $\frac{\pi}{2} - 3x = 2n\pi - x; n \in I$

$\Rightarrow 4x = \frac{\pi}{2} - 2n\pi; n \in I$ or $2x = \frac{\pi}{2} - 2n\pi; n \in I$

$\Rightarrow x = \frac{\pi}{8} - \frac{n\pi}{2}; n \in I$ or $x = \frac{\pi}{4} - n\pi; n \in I$

$\because x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\Rightarrow x = \frac{\pi}{8}, -\frac{3\pi}{8}, \frac{\pi}{4}$

$\Rightarrow y = \cos\frac{\pi}{8}, \cos\frac{3\pi}{8}, \frac{1}{\sqrt{2}}$

$\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right), \left(\frac{\pi}{8}, \cos\frac{\pi}{8}\right), \left(-\frac{3\pi}{8}, \cos\frac{3\pi}{8}\right)$

2. Find the values of $x \in (-\pi, \pi)$ which satisfy the equation, $8^{\left(1 + |\cos x| + |\cos x|^2 + |\cos x|^3 + .....\right)} = 4^3$.

अन्तराल $x \in (-\pi, \pi)$ में x का मान ज्ञात करो जो निम्न समीकरण $8^{\left(1 + |\cos x| + |\cos x|^2 + |\cos x|^3 + .....\right)} = 4^3$ को सन्तुष्ट कर सके।

**Ans.** $\pm \frac{\pi}{3}$ , $\pm \frac{2\pi}{3}$

**Sol.** Let $|\cos x| = t$

$\Rightarrow 8^{\left(1+|\cos x|+|\cos x|^2+.....\right)} = 4^3 = 2^6 \Rightarrow 8^{\left(1+t+t^2+.....\right)} = 2^6$

$\Rightarrow 8^{\frac{1}{1-t}} = 2^6$ $\qquad$ $\Rightarrow 2^{\frac{3}{1-t}} = 2^6$

$\Rightarrow \frac{3}{6} = 1-t$ $\qquad$ $\Rightarrow t = \frac{1}{2}$

$\Rightarrow |\cos x| = \frac{1}{2}$

$\because\ x \in (-\pi, \pi)$

$\Rightarrow x = \pm\frac{\pi}{3}, \pm\frac{2\pi}{3}$

3. Solve the following system of equations for x and y

$5^{\left(\cos ec^2 x - 3\sec^2 y\right)} = 1$ and $2^{\left(2\cos ec\, x - \sqrt{3}\,|\sec y|\right)} = 64$

x व y के लिए निम्न समीकरण निकाय का हल ज्ञात करे

$5^{\left(\cos ec^2 x - 3\sec^2 y\right)} = 1$ and $2^{\left(2\cos ec\, x - \sqrt{3}\,|\sec y|\right)} = 64.$

**Ans.** $x = k\pi + (-1)^n \left(\sin^{-1}\frac{1}{6}\right)$ , $y = t\,\pi \pm \left(\cos^{-1}\frac{1}{\sqrt{12}}\right)$

**Sol.** $5^{\cos ec^2 x - 3\sec^2 y} = 1$

$\Rightarrow \text{cosec}^2 x - 3\ \sec^2 y = 0$

$\Rightarrow \left(\cos ec x + \sqrt{3}\sec y\right)\left(\cos ec x - \sqrt{3}\sec y\right) = 0$

$\Rightarrow \cos ec x + \sqrt{3}\sec y = 0$ $\qquad$ or $\qquad$ $\cos ec x - \sqrt{3}\sec y = 0$

Now $2^{\left(2\cos ec x - \sqrt{3}|\sec y|\right)} = 64 = 2^6$

$\Rightarrow 2\cos ec x - \sqrt{3}\,|\sec y| = 6$

If $\sec y > 0$ then $2\cos ec x - \sqrt{3}\sec y = 6$

If $\sec y < 0$ then $2\cos ec x + \sqrt{3}\sec y = 6$

**Case I** :- If $\cos ec x - \sqrt{3}\sec y = 0$ then $\cos ec x = \sqrt{3}\sec y$

$\Rightarrow 2\sqrt{3}\sec y - \sqrt{3}\,|\sec y| = 6$

$\Rightarrow 2\sec y - |\sec y| = \frac{6}{\sqrt{3}} = 2\sqrt{3}$

$\Rightarrow \sec y = 2\sqrt{3}$ or $\sec y = \frac{2}{\sqrt{3}}$ (If $\sec y < 0$)

$\Rightarrow \sec y = \sqrt{12}$

$\Rightarrow \cos y = \frac{1}{\sqrt{12}}$

$\Rightarrow y = 2n\pi \pm \cos^{-1}\left(\frac{1}{\sqrt{12}}\right)$

& $x = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{6}\right)$

**Case II** :- If $\text{cosec}\, x + \sqrt{3}\sec y = 0$ & $2\sqrt{3}\sec y - \sqrt{3}|\sec y| = 6$

If $\sec y > 0$ then $\sec y = -\frac{6}{3\sqrt{3}} < 0$ (Not possible)

If $\sec y < 0$ then $\sec y = -\sqrt{12}$

$\Rightarrow \cos y = -\frac{1}{\sqrt{12}}$

$\Rightarrow y = 2n\pi \pm \cos^{-1}\left(\frac{1}{\sqrt{12}}\right)$ & $x = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{6}\right)$

4. Find the value of $\theta$, which satisfy $3 - 2\cos\theta - 4\sin\theta - \cos 2\theta + \sin 2\theta = 0$.

समीकरण $3 - 2\cos\theta - 4\sin\theta - \cos 2\theta + \sin 2\theta = 0$ को हल कीजिए।

**Ans.** $\theta = 2n\pi$ or $2n\pi + \frac{\pi}{2}$ ; $n \in I$

**Sol.** $3 - 2\cos\theta - 4\sin\theta - \cos 2\theta + \sin 2\theta = 0$

$\Rightarrow 3 - 2\cos\theta - 4\sin\theta - (-2\sin^2\theta + 1) + 2\sin\theta\cos\theta = 0$

$\Rightarrow 3 + 2\cos\theta(\sin\theta - 1) - 4\sin\theta + 2\sin^2\theta - 1 = 0$

$\Rightarrow 2\sin^2\theta - 4\sin\theta + 2 + 2\cos\theta(\sin\theta - 1) = 0$

$\Rightarrow 2(\sin\theta - 1)^2 + 2\cos\theta(\sin\theta - 1) = 0$

$\Rightarrow (\sin\theta - 1)(\sin\theta + \cos\theta - 1) = 0$

$\Rightarrow \sin\theta = 1$ or $\sin\theta + \cos\theta = 1$

$\Rightarrow \sin\theta = 1$ or $\cos\left(\theta - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\Rightarrow \theta = (4n+1)\frac{\pi}{2}; n \in I$ or $\theta - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}; n \in I$

$\Rightarrow \theta = 2n\pi + \frac{\pi}{2}; n \in I$ or $\theta = 2n\pi; n \in I$

5. Find the least positive angle measured in degrees satisfying the equation $\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$.

न्यूनतम धनात्मक कोण जो समीकरण $\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$ को संतुष्ट करता हो, का डिग्री में मान ज्ञात कीजिए।

**Ans.** 72°

**Sol.** Given $\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$

$$\left.\begin{array}{l} \text{If } a^3 + b^3 + c^3 = (a+b+c)^3 \\ \text{then } 3(a+b)(b+c)(c+a) = 0 \end{array}\right\}$$

$\Rightarrow (\sin x + \sin 2x)(\sin 2x + \sin 3x)(\sin 3x + \sin x) = 0$

$\Rightarrow 8 \sin\frac{3x}{2} \cos^2\frac{x}{2} \cdot \sin 3x \cdot \cos x \cdot \sin\frac{5x}{2} = 0$

$\Rightarrow \sin 3x = 0$ or $\cos x = 0$ or $\cos\frac{x}{2} = 0$

or $\sin\frac{3x}{2} = 0$ or $\sin\frac{5x}{2} = 0$

$\Rightarrow x = 0, 3\pi, \frac{\pi}{2}, \frac{3\pi}{2}, \pi, \frac{2\pi}{3}, \frac{2\pi}{5}$

$\Rightarrow$ Smallest positive value of x is $\frac{2\pi}{5} = 72^\circ$

6. Prove that the equations :

(a) $\sin x \cdot \sin 2x \cdot \sin 3x = 1$ (b) $\sin x \cdot \cos 4x \cdot \sin 5x = -1/2$

have no solution.

सिद्ध कीजिए की निम्न समीकरणों का कोई हल नहीं होगा :

(a) $\sin x \cdot \sin 2x \cdot \sin 3x = 1$ (b) $\sin x \cdot \cos 4x \cdot \sin 5x = -1/2$

**Ans.**

**Sol.** (a) $\sin x \cdot \sin 2x \cdot \sin 3x = 1$

$\because \sin x \in [-1, 1]$

$\sin 2x \in [-1, 1]$

$\sin 3x \in [-1, 1]$

This is possible only when

(i) $\sin x = 1$ & $\sin 2x = 1$ & $\sin 3x = 1$

This is not possible.

(ii) $\sin x = -1$ & $\sin 2x = 1$ & $\sin 3x = -1$

This is not possible.

$\Rightarrow \sin x \cdot \sin 2x \cdot \sin 3x = 1$

Equation has no solution.

(b) $\sin x \cdot \cos 4x \cdot \sin 5x = -\frac{1}{2}$

$\Rightarrow (2 \sin x \sin 5x)(\cos 4x) = -1$

$\Rightarrow (\cos 4x - \cos 6x)\cos 4x = -1$

$\Rightarrow \cos^2 4x + 1 = \cos 6x \cos 4x$

$\Rightarrow 2(\cos^2 4x + 1) = 2\cos 6x \cos 4x = \cos 10x + \cos 2x$

$\cos 10x + \cos 2x \in [-2, 2]$ & $2(\cos^2 x + 1) \in [2, 4]$

$\therefore$ Only possibility are $\cos^2 4x + 1 = 1$ & $\cos 10x + \cos 2x = 2$

$\Rightarrow \cos^2 4x = 0$ & $\cos 10x = 1$ & $\cos 2x = 1$

$\because \cos^2 2x = \frac{1+\cos 4x}{2} = \frac{1}{2} \Rightarrow \cos 2x = \frac{\pm 1}{\sqrt{2}}$

$\therefore \sin x \cdot \cos 4x \cdot \sin 5x = \frac{-1}{2}$ have no solution.

7. Solve the equation : $\sin 5x = 16 \sin^5 x$.

समीकरण : $\sin 5x = 16 \sin^5 x$ को हल करें।

**Ans.** $x = n\pi$ or $x = n\pi \pm \frac{\pi}{6}$

**Sol.** $\sin 5x = 16 \sin^5 x$

$\Rightarrow \sin(3x + 2x) = 16 \sin^5 x$

$\Rightarrow \sin 3x \cos 2x + \cos 3x \sin 2x = 16 \sin^5 x$

$\Rightarrow (3 \sin x - 4\sin^3 x)(1 - 2\sin^2 x) + (4\cos^3 x - 3\cos x) 2 \sin x \cos x = 16 \sin^5 x$

$\Rightarrow 3 \sin x - 10 \sin^3 x + 8 \sin^5 x + 8 \sin x \cos^4 x - 6 \sin x \cos^2 x = 16 \sin^5 x$

$\Rightarrow 3 \sin x - 10 \sin^3 x + 8 \sin^5 x - 6 \sin x + 8 \sin x + 8 \sin^5 x - 16 \sin^3 x + 6 \sin^3 x = 16 \sin^5 x$

$\Rightarrow 5 \sin x - 20 \sin^3 x = 0$

$\Rightarrow \sin x(1 - 4\sin^2 x) = 0$

$\Rightarrow \sin x = 0$ or $\sin^2 x = \frac{1}{4} = \sin^2 \frac{\pi}{6}$

$\Rightarrow x = n\pi;\ n \in I$ or $x = n\pi \pm \frac{\pi}{6};\ n \in I$

8. Find all the solutions of, $4\cos^2x\ \sin x - 2\sin^2x = 3\sin x$.

$4\cos^2x\ \sin x - 2\sin^2x = 3\sin x$ के सभी हल निकालिए।

**Ans.** $n\pi\ ;\ n\pi + (-1)^n\ \frac{\pi}{10}$ or $n\pi + (-1)^n\left(-\frac{3\pi}{10}\right)$

**Sol.** $4\cos^2x \sin x - 2\sin^2x = 3\sin x$

$\Rightarrow 4(1 - \sin^2x)\sin x - 2\sin^2x = 3\sin x$

$\Rightarrow 4\sin x - 4\sin^3x - 2\sin^2x - 3\sin x = 0$

$\Rightarrow \sin x(1 - 2\sin x - 4\sin^2x) = 0$

$\Rightarrow \sin x = 0$ or $4\sin^2x + 2\sin x - 1 = 0$

$\Rightarrow \sin x = 0$ or $\sin x = \frac{\sqrt{5}-1}{4}$ or $\sin x = \frac{-\left(-1+\sqrt{5}\right)}{4}$

$\Rightarrow x = n\pi;\ n \in I$ or $x = n\pi + (-1)^n\frac{\pi}{10};\ n \in I$ or $x = n\pi + (-1)^n\left(-\frac{3\pi}{10}\right)$

9. Solve for $x$, $(-\pi \le x \le \pi)$ the equation; $2(\cos x + \cos 2x) + \sin 2x\,(1 + 2\cos x) = 2\sin x$.

$x$, $(-\pi \le x \le \pi)$ के मानों के लिये समीकरण $2(\cos x + \cos 2x) + \sin 2x\,(1 + 2\cos x) = 2\sin x$ को हल कीजिए।

**Ans.** $\frac{\pm\pi}{3}, \frac{-\pi}{2}, \pm\pi$

**Sol.** $2(\cos x + \cos 2x) + \sin 2x\,(1 + 2\cos x) = 2\sin x$

$\Rightarrow 2(\cos x + 1 - 2\sin^2x) + 2\sin x\cos x\,(1 + 2\cos x) = 2\sin x$

$\Rightarrow 2\cos x + 2 - 4\sin^2x + 2\sin x\cos x + 4\sin x\cos^2x - 2\sin x = 0$

$\Rightarrow 2\cos x + 2\sin x\cos x - 2\sin x + 2 - 4\sin^2x + 4\sin x - 4\sin^3x = 0$

$\Rightarrow 2\cos x + 2\sin x(1 + \cos x) + 2 - 4\sin^2x - 4\sin^3x = 0$

$\Rightarrow 2\cos x(1 + \sin x) + 2(\sin x + 1) - 4\sin^2x(1 + \sin x) = 0$

$\Rightarrow (1 + \sin x)(2\cos x + 2 - 4\sin^2x) = 0$

$\Rightarrow (\sin x + 1)(\cos x + 1 - 2(1 - \cos^2x)) = 0$

$\Rightarrow (\sin x + 1)(2\cos^2x + \cos x - 1) = 0$

$\Rightarrow \sin x = -1$ or $\cos x = -1$ or $\cos x = \frac{1}{2}$

$\Rightarrow \sin x = -1$ or $\cos x = -1$ or $\cos x = -\frac{1}{2}$

$\Rightarrow x = \pm\frac{\pi}{3}, \frac{-\pi}{2}, \pm\pi$

10. Find the general solution of the equation $2(\sin x - \cos 2x) - \sin 2x(1 + 2\sin x) + 2\cos x = 0$.

समीकरण $2(\sin x - \cos 2x) - \sin 2x(1 + 2\sin x) + 2\cos x = 0$ का व्यापक हल बताइए।

**Ans.** $x = 2n\pi$ or $x = n\pi + (-1)^n\left(-\frac{\pi}{2}\right)$ or $x = n\pi + (-1)^n\frac{\pi}{6}$

**Sol.** $2(\sin x - \cos 2x) - \sin 2x(1 + 2\sin x) + 2\cos x = 0$

$\Rightarrow 2\sin x - 2(1 - 2\sin^2 x) - 2\sin x\cos x(1 + 2\sin x) + 2\cos x = 0$

$\Rightarrow 2\sin x - 2 + 4\sin^2 x - 2\sin x\cos x - 4\sin^2 x\cos x + 2\cos x = 0$

$\Rightarrow 2\sin x(1 - \cos x) + 4\sin^2 x(1 - \cos x) - 2(1 - \cos x) = 0$

$\Rightarrow (1 - \cos x)(2\sin x + 4\sin^2 x - 2) = 0$

$\Rightarrow (1 - \cos x)(\sin x + 2\sin^2 x - 1) = 0$

$\Rightarrow \cos x = 1$ or $\sin x = -1$ or $\sin x = \frac{1}{2}$

$\Rightarrow x = 2n\pi;\ n \in I$ or $x = n\pi + (-1)^n\left(-\frac{\pi}{2}\right)$ or $x = n\pi + (-1)^n\frac{\pi}{6}$

11. Find the values of x, between 0 and $2\pi$, satisfying the equation $\cos 3x + \cos 2x = \sin\frac{3x}{2} + \sin\frac{x}{2}$.

0 से $2\pi$ में समीकरण $\cos 3x + \cos 2x = \sin\frac{3x}{2} + \sin\frac{x}{2}$ को संतुष्ट करने वाले x के मान ज्ञात कीजिए।

**Ans.** $\frac{\pi}{7}, \frac{5\pi}{7}, \pi, \frac{9\pi}{7}, \frac{13\pi}{7}$

**Sol.** $\cos 3x + \cos 2x = \sin\frac{3x}{2} + \sin\frac{x}{2}$

$\Rightarrow \cos\frac{5x}{2}\cos\frac{x}{2} = \sin(x)\cos\frac{x}{2}$

$\Rightarrow \cos\frac{x}{2}\left(\cos\frac{5x}{2} - \sin x\right) = 0$

$\Rightarrow \cos\frac{x}{2} = 0$ or $\cos\left(\frac{5x}{2}\right) = \cos\left(\frac{\pi}{2} - x\right)$

$\Rightarrow \cos\frac{x}{2} = 0$ or $\frac{5x}{2} = 2n\pi \pm \frac{\pi}{2} - x$

$\because\ 0 < x < 2\pi$

$\Rightarrow \frac{x}{2} = \frac{\pi}{2}$ or $x = \frac{\pi}{7}, \frac{9\pi}{7}, \frac{13\pi}{7}, \pi, \frac{5\pi}{7}$

$\Rightarrow x = \frac{\pi}{7}, \frac{5\pi}{7}, \pi, \frac{9\pi}{7}, \frac{13\pi}{7}$

12. Solve: $\tan^2 2x + \cot^2 2x + 2\tan 2x + 2\cot 2x = 6$.

$\tan^2 2x + \cot^2 2x + 2\tan 2x + 2\cot 2x = 6$ को हल करें।

**Ans.** $x = \frac{n\pi}{4} + (-1)^n \frac{\pi}{8}$ or $\frac{n\pi}{4} + (-1)^{n+1} \frac{\pi}{24}$

**Sol.** $\tan^2 2x + \cot^2 2x + 2\tan 2x + 2\cot 2x = 6$

$\Rightarrow (\tan 2x + \cot 2x)^2 + 2(\tan 2x + \cot 2x) = 8$

Let $\tan 2x + \cot 2x = t$

$\Rightarrow t^2 + 2t - 8 = 0$

$\Rightarrow t = 2$ or $t = -4$

$\Rightarrow \tan^2 2x - 2\tan 2x + 1 = 0$ or $\tan^2 2x + 4\tan 2x + 1 = 0$

$\Rightarrow \tan 2x = 1$ or $\tan^2 2x + 4\tan 2x + 1 = 0$

$\Rightarrow 2x = n\pi + \frac{\pi}{4}$ or $\frac{\sin^2 2x}{\cos^2 2x} + 1 = \frac{-4\sin 2x}{\cos 2x}$

$\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}; n \in I$ or $\frac{1}{\cos^2 2x} = \frac{-4\sin 2x}{\cos 2x}$

or $\sin 4x = \frac{-1}{2} = \sin\left(-\frac{\pi}{6}\right)$

$\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}; n \in I$ or $x = \frac{n\pi}{4} + (-1)^{n+1} \frac{\pi}{24}$

13. Solve the equation: $1 + 2\operatorname{cosec} x = -\frac{\sec^2 \frac{x}{2}}{2}$.

$1 + 2\operatorname{cosec} x = -\frac{\sec^2 \frac{x}{2}}{2}$ को हल करें।

**Ans.** $x = 2n\pi - \frac{\pi}{2}$

**Sol.** Let $\tan \frac{x}{2} = t$

$\Rightarrow \sin x = \frac{2t}{1+t^2}$ & $\sec^2 \frac{x}{2} = 1 + t^2$

$\Rightarrow 1 + \frac{2(1+t^2)}{2t} + \frac{1}{2}(1+t^2) = 0$

$$\Rightarrow 1+\frac{1+t^2}{t}+\frac{1}{2}+\frac{t^2}{2}=0$$

$\Rightarrow t^3 + t + 2 + 2t^2 + 2t = 0$

$\Rightarrow t^3 + 2t^2 + 3t + 2 = 0$

$\Rightarrow (t + 1)(t^2 + t + 2) = 0$

$\Rightarrow t = -1$

$$\Rightarrow \tan\frac{x}{2}=\tan\left(-\frac{\pi}{4}\right)$$

$$\Rightarrow \frac{x}{2}=n\pi-\frac{\pi}{4}$$

$$\Rightarrow x=2n\pi-\frac{\pi}{2};n\in I$$

14. Solve: $\tan^2x \cdot \tan^2 3x \cdot \tan 4x = \tan^2x - \tan^2 3x + \tan 4x$.

$\tan^2x \cdot \tan^2 3x \cdot \tan 4x = \tan^2x - \tan^2 3x + \tan 4x$ को हल करें।

**Ans.** $\frac{(2n+1)\pi}{4}$ , $k\pi$ , where $n, k \in I$

**Sol.** $\tan4x\,[(\tan x \tan 3x)^2 - 1] = \tan^2x - \tan^2 3x$

$$\Rightarrow \tan 4x=\frac{\tan^2 x-\tan^2 3x}{\tan^2 x\tan^2 3x-1}=\frac{\tan^2 3x-\tan^2 x}{1-\tan^2 x\tan^2 3x}=\tan(3x-x)\cdot\tan(3x+x)$$

$\Rightarrow \tan4x = \tan2x \tan4x$

$\Rightarrow \tan4x = 0$ or $\tan2x = 1$ (Not possible as tan4x is not defined)

$$\Rightarrow x=\frac{n\pi}{4}$$

$$\Rightarrow x=\frac{n\pi}{4};n\in I$$

$$\because x\neq(2n+1)\frac{\pi}{2}$$

$$\Rightarrow x=\frac{(2n+1)\pi}{4};n\in I \quad \text{or} \quad x = k\pi;\ k \in I$$

15. Find the set of values of x satisfying the equality $\sin\left(x-\frac{\pi}{4}\right) - \cos\left(x+\frac{3\pi}{4}\right) = 1$ and the inequality

$$\frac{2\cos 7x}{\cos 3+\sin 3}>2^{\cos 2x}.$$

समीकरण $\sin\left(x-\frac{\pi}{4}\right)-\cos\left(x+\frac{3\pi}{4}\right)=1$ तथा असमिका $\frac{2\cos 7x}{\cos 3+\sin 3}>2^{\cos 2x}$ को सन्तुष्ट करने वाले x के मान ज्ञात करें।

**Ans.** $x=2n\pi+\frac{3\pi}{4}$, $n\in I$

**Sol.** $\sin\left(x-\frac{\pi}{4}\right)-\cos\left(x+\frac{3\pi}{4}\right)=1$

$\Rightarrow \sin x\cos\frac{\pi}{4}-\cos x\cos\frac{\pi}{4}-\cos x\cos\frac{3\pi}{4}+\sin x\sin\frac{3\pi}{4}=1$

$\Rightarrow \sqrt{2}\sin x=1$

$\Rightarrow \sin x=\frac{1}{\sqrt{2}}\Rightarrow x=\frac{\pi}{4},\frac{3\pi}{4}$

$\Rightarrow \cos 2x=1-2\sin^2 x=1-\frac{2}{2}=1-1=0$

$\Rightarrow \frac{2\cos 7x}{\cos x3+\sin 3}>2^{\cos 2x}$

$\Rightarrow \frac{2\cos 7x}{\cos 3+\sin 3}>1$

$\because \cos 3+\sin 3<0$

$\Rightarrow \cos 7x<0$ If $x=\frac{3\pi}{4}$

$\therefore$ equation has solution.

$\Rightarrow$ General solution is $x=2n\pi+\frac{3\pi}{4};n\in I$

16. Solve: $\sin\left(\frac{\sqrt{x}}{2}\right)+\cos\left(\frac{\sqrt{x}}{2}\right)=\sqrt{2}\sin\sqrt{x}$.

$\sin\left(\frac{\sqrt{x}}{2}\right)+\cos\left(\frac{\sqrt{x}}{2}\right)=\sqrt{2}\sin\sqrt{x}$ को हल करें।

**Ans.** $x=\left(4n\pi+\frac{\pi}{2}\right)^2$ or $x=\left(\frac{4m\pi}{3}+\frac{\pi}{2}\right)^2$ where $m,n\in W$.

**Sol.** $\sin\left(\frac{\sqrt{x}}{2}\right)+\cos\left(\frac{\sqrt{x}}{2}\right)=\sqrt{2}\sin\sqrt{x}$

$\Rightarrow \frac{1}{\sqrt{2}}\sin\left(\frac{\sqrt{x}}{2}\right)+\frac{1}{\sqrt{2}}\cos\left(\frac{\sqrt{x}}{2}\right)=\sin\sqrt{x}$

For $\sqrt{x}$ ; $x \geq 0$

$\Rightarrow \sin\left(\frac{\pi}{4}+\frac{\sqrt{x}}{2}\right)=\sin\sqrt{x}$

$\Rightarrow \sqrt{x}=n\pi+(-1)^n\left(\frac{\pi}{4}+\frac{\sqrt{x}}{2}\right)$

$\Rightarrow \sqrt{x}-\frac{\sqrt{x}}{2}=2n\pi+\frac{\pi}{4}$ or $\sqrt{x}+\frac{\sqrt{x}}{2}=(2m+1)\pi-\frac{\pi}{4}$

$\Rightarrow \sqrt{x}=4n\pi+\frac{\pi}{2}$ or $\sqrt{x}=\frac{4m\pi}{3}+\frac{\pi}{2}$

$\Rightarrow x=\left(4n\pi+\frac{\pi}{2}\right)^2 ; n \in W$ or $x=\left(\frac{4m\pi}{3}+\frac{\pi}{2}\right)^2 ; m \in W$

17. Solve for x and y, $\begin{matrix} x\cos^3 y + 3x\cos y \sin^2 y = 14 \\ x\sin^3 y + 3x\cos^2 y \sin y = 13 \end{matrix}$.

समीकरणों $\begin{matrix} x\cos^3 y + 3x\cos y \sin^2 y = 14 \\ x\sin^3 y + 3x\cos^2 y \sin y = 13 \end{matrix}$ के x, y के लिये हल करें।

**Ans.** $x = \pm 5\sqrt{5}$ & $y = n\pi + \tan^{-1}\frac{1}{2}$

**Sol.** $x\cos^3 y + 3x\cos y \sin^2 y = 14$ ...(i)

$x\sin^3 y + 3x\cos^2 y \sin y = 13$ ...(ii)

By (i) + (ii); we get

$x(\cos^3 y + \sin^3 y) + 3x \sin y \cos y (\sin y + \cos y) = 27$

$\Rightarrow x((\cos y + \sin y)^3 - 3\cos y \sin y (\cos y + \sin y)) + 3x \sin y \cos y (\sin y + \cos y) = 27$

$\Rightarrow x(\cos y + \sin y)^3 = 27$

$\Rightarrow x = \left(\frac{3}{\cos y + \sin y}\right)^3$ ...(iii)

By (i) – (ii); we get

$x(\cos^3 y - \sin^3 y) + 3x \sin y \cos y (\sin y - \cos y) = 1$

$\Rightarrow x(\cos y - \sin y)^3 = 1$

$$\Rightarrow x = \frac{1}{(\cos y - \sin y)^3} \qquad ...(iv)$$

By (iii) & (iv)

$$\frac{\cos y + \sin y}{\cos y - \sin y} = \frac{3}{1}$$

Apply C & D Rule

$$\tan y = \frac{1}{2}$$

$$\Rightarrow y = n\pi + \tan^{-1}\left(\frac{1}{2}\right); n \in I$$

$$\Rightarrow x = \pm 5\sqrt{5}$$

18. Let $A = \left\{\theta \in R \,\middle|\, \cos^2(\sin\theta) + \sin^2(\cos\theta) = 1\right\}$ and $B = \left\{\theta \in R \,\middle|\, \cos(\sin\theta) \cdot \sin(\cos\theta) = 0\right\}$, then find the number of elements in the set $A \cap B$.

माना $A = \left\{\theta \in R \,\middle|\, \cos^2(\sin\theta) + \sin^2(\cos\theta) = 1\right\}$ तथा $B = \left\{\theta \in R \,\middle|\, \cos(\sin\theta) \cdot \sin(\cos\theta) = 0\right\}$ है, तो समुच्चय $A \cap B$ में अवयवों की संख्या ज्ञात कीजिये।

**Ans.** 0

**Sol.** $A = \{\theta \in R \mid \cos^2(\sin\theta) + \sin^2(\cos\theta) = 1\}$

$B = \{\theta \in R \mid \cos(\sin\theta) \cdot \sin(\cos\theta) = 0\}$

If $\theta \in A \cap B$

then

$\cos^2(\sin\theta) + \sin^2(\cos\theta) = 1$ & $\cos(\sin\theta) \cdot \sin(\cos\theta) = 0$

If $\cos(\sin\theta) = 0$

Then $\underbrace{\sin\theta = (2n+1)\frac{\pi}{2}; n \in I}_{\text{Not possible}}$

If $\sin(\cos\theta) = 0$ then $\cos\theta = n\pi$; $n \in I$

Only possibility are $\cos\theta = 0$ but $\cos^2(1) \neq 1$

$\therefore$ No. of element in $A \cap B$ is 0.

## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question 19 to 21**

Let $S_1$ be the set of all those solutions of the equation $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$ which are independent of a and b and $S_2$ be the set of all such solutions which are dependent on a and b. Then

माना $S_1$ समीकरण $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$ के उन सभी हलों का समुच्चय है जो a तथा b से स्वतंत्र हैं तथा $S_2$ उन सभी हलों का समुच्चय है जो a तथा b पर निर्भर करते हैं।

19. The set $S_1$ and $S_2$ are :

(A) $\{n\pi;\ n \in I\}$ and $\left\{\frac{n\pi}{2} + \frac{b}{2} + (-1)^n \frac{\sin^{-1}(a\sin b)}{2};\ n \in I\right\}$

(B) $\{n\pi;\ n \in I\}$ and $\left\{\frac{n\pi}{2} + (-1)^n \frac{\sin^{-1}(b\sin a)}{2};\ n \in I\right\}$

(C) $\left\{\frac{n\pi}{2};\ \ n \in I\right\}$ and $\left\{\frac{n\pi}{2} + \frac{b}{2} + (-1)^n \sin^{-1}\left(\frac{a}{2}\sin b\right);\ n \in I\right\}$

(D) none of these

समुच्चय $S_1$ तथा $S_2$ है :

(A) $\{n\pi;\ n \in I\}$ तथा $\left\{\frac{n\pi}{2} + \frac{b}{2} + (-1)^n \frac{\sin^{-1}(a\sin b)}{2};\ n \in I\right\}$

(B) $\{n\pi;\ n \in I\}$ तथा $\left\{\frac{n\pi}{2} + (-1)^n \frac{\sin^{-1}(b\sin a)}{2};\ n \in I\right\}$

(C) $\left\{\frac{n\pi}{2};\ \ n \in I\right\}$ तथा $\left\{\frac{n\pi}{2} + \frac{b}{2} + (-1)^n \sin^{-1}\left(\frac{a}{2}\sin b\right);\ n \in I\right\}$

(D) इनमें से कोई नहीं

**Ans.** A

**Sol.** $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$

$\Rightarrow \cos\theta\cos(2\theta - b) + a\cos\theta\cos(2\theta - b) = \cos(\theta - b) + a\cos 2\theta\cos(\theta - b)$

$\Rightarrow 2\cos\theta\cos(2\theta - b) + 2a\cos\theta\cos(2\theta - b)$

$= 2\cos(\theta - b) + 2a\cos 2\theta\cos(\theta - b)$

$\Rightarrow \cos(3\theta - b) + \cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta - b))$

$= 2\cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta + b))$

$\Rightarrow \cos(3\theta - b) + a\cos(\theta - b) = \cos(\theta - b) + a\cos(\theta + b)$

$\Rightarrow \cos(3\theta - b) - \cos(\theta - b) = a(\cos(\theta + b) - \cos(\theta - b))$

$\Rightarrow$ $2\sin(2\theta - b)\sin\theta = 2a\sin\theta\sin b$

$\Rightarrow$ $\sin\theta(\sin(2\theta - b) - a\sin b) = 0$

$\Rightarrow$ $\sin\theta = 0$ or $\sin(2\theta - b) = a\sin b$

$\Rightarrow$ $\sin\theta = 0$ $\Rightarrow \theta = n\pi$, $n \in I$.

and $\sin(2\theta - b) = a\sin b$.

$\Rightarrow$ $\sin(2\theta - b) = \sin(\sin^{-1}(a\sin b))$

$\Rightarrow$ $2\theta - b = n\pi + (-1)^n \sin^{-1}(a\sin b)$ $n \in I$

$S_1 = n\pi$, $n \in I$.

$S_2 = \frac{1}{2}(n\pi + (-1)^n \sin^{-1}(a\sin b) + b)$ $n \in I$.

20. Conditions that should be imposed on a and b such that $S_2$ is non-empty :

(A) $\left|\frac{a}{2}\sin b\right| < 1$ (B) $\left|\frac{a}{2}\sin b\right| \le 1$ (C) $|a\sin b| \le 1$ (D) none of these

a तथा b पर शर्त (condition) लगानी चाहिये ताकि $S_2$ अरिक्त हो :

(A) $\left|\frac{a}{2}\sin b\right| < 1$ (B) $\left|\frac{a}{2}\sin b\right| \le 1$ (C) $|a\sin b| \le 1$ (D) इनमें से कोई नहीं

**Ans.** C

**Sol.** $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos2\theta)\cos(\theta - b)$

$\Rightarrow$ $\cos\theta\cos(2\theta - b) + a\cos\theta\cos(2\theta - b) = \cos(\theta - b) + a\cos2\theta\cos(\theta - b)$

$\Rightarrow$ $2\cos\theta\cos(2\theta - b) + 2a\cos\theta\cos(2\theta - b)$

$= 2\cos(\theta - b) + 2a\cos2\theta\cos(\theta - b)$

$\Rightarrow$ $\cos(3\theta - b) + \cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta - b))$

$= 2\cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta + b))$

$\Rightarrow$ $\cos(3\theta - b) + a\cos(\theta - b) = \cos(\theta - b) + a\cos(\theta + b)$

$\Rightarrow$ $\cos(3\theta - b) - \cos(\theta - b) = a(\cos(\theta + b) - \cos(\theta - b))$

$\Rightarrow$ $2\sin(2\theta - b)\sin\theta = 2a\sin\theta\sin b$

$\Rightarrow$ $\sin\theta(\sin(2\theta - b) - a\sin b) = 0$

$\Rightarrow$ $\sin\theta = 0$ or $\sin(2\theta - b) = a\sin b$

$\Rightarrow$ $\sin\theta = 0$ $\Rightarrow \theta = n\pi$, $n \in I$.

and $\sin(2\theta - b) = a\sin b$.

$\Rightarrow$ $\sin(2\theta - b) = \sin(\sin^{-1}(a\sin b))$

$\Rightarrow$ $2\theta - b = n\pi + (-1)^n \sin^{-1}(a\sin b)$ $n \in I$

$|a\sin b| \le 1 \rightarrow$ For $S_2$ non empty.

21. All the permissible values of b, if $a = 0$ and $S_2 \cap (0, \pi)$ is non empty, is :

(A) $b \in (-n\pi, 2n\pi)$, $n \in I$ (B) $b \in (-n\pi, 2\pi - n\pi)$, $n \in I$

(C) $b \in (-n\pi, n\pi)$ (D) none of these

b के सभी अनुमेय मान, यदि $a = 0$ तथा $S_2 \cap (0, \pi)$ अरिक्त हो, होंगे :

(A) $b \in (-n\pi, 2n\pi)$, $n \in I$ (B) $b \in (-n\pi, 2\pi - n\pi)$, $n \in I$

(C) $b \in (-n\pi, n\pi)$ (D) इनमें से कोई नहीं

**Ans.** B

**Sol.** $(1 + a)\cos\theta \cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$

$\Rightarrow \cos\theta \cos(2\theta - b) + a\cos\theta\cos(2\theta - b) = \cos(\theta - b) + a\cos 2\theta \cos(\theta - b)$

$\Rightarrow 2\cos\theta\cos(2\theta - b) + 2a\cos\theta\cos(2\theta - b)$

$= 2\cos(\theta - b) + 2a\cos 2\theta\cos(\theta - b)$

$\Rightarrow \cos(3\theta - b) + \cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta - b))$

$= 2\cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta + b))$

$\Rightarrow \cos(3\theta - b) + a\cos(\theta - b) = \cos(\theta - b) + a\cos(\theta + b)$

$\Rightarrow \cos(3\theta - b) - \cos(\theta - b) = a(\cos(\theta + b) - \cos(\theta - b))$

$\Rightarrow 2\sin(2\theta - b)\sin\theta = 2a\sin\theta\sin b$

$\Rightarrow \sin\theta(\sin(2\theta - b) - a\sin b) = 0$

$\Rightarrow \sin\theta = 0$ or $\sin(2\theta - b) = a\sin b$

$\Rightarrow \sin\theta = 0 \Rightarrow \theta = n\pi, \quad n \in I.$

and $\sin(2\theta - b) = a\sin b.$

$\Rightarrow \sin(2\theta - b) = \sin(\sin^{-1}(a\sin b))$

$\Rightarrow 2\theta - b = n\pi + (-1)^n \sin^{-1}(a\sin b)$ $n \in I$

If $a = 0$

$\sin(2\theta - b) = 0$

$2\theta - b = n\pi$ $n \in I$

for $S_2$ a subset of $(0,\pi)$

$$0 < \frac{n\pi + b}{2} < \pi \qquad n \in I$$

$\Rightarrow -n\pi < b < 2\pi - n\pi.$

$b \in (-n\pi, 2\pi - n\pi)$, $n \in I$.

## MATCH THE COLUMN TYPE QUESTION(S)

22. Match the following column :

| **Column-I (Equation)** | **Column-II (General Solution)** |
|---|---|
| (A) $\frac{1 - \cos 2x}{\sin 2x} = 0$ | (P) $(8n - 1)\frac{\pi}{4}$, $n \in I$ |
| (B) $\cos^2 x + 5\cos x = 2\sin^2 x$ | (Q) $2n\pi \pm \cos^{-1}\frac{1}{3}$, $n \in I$ |
| (C) $\sin^4 x + \sin^4\left(\frac{x}{2} + \frac{\pi}{8}\right) + \cos^4 x = \frac{1}{2}\sin^2 2x$ | (R) No solution |
| (D) $4\cos^2\theta \cdot \sin\theta - 2\sin^2\theta = 3\sin\theta$ | (S) $n\pi + (-1)^n\left(-\frac{3\pi}{10}\right)$, $n \in I$ |

निम्न कॉलम को सुमेलित कीजिए :

| **Column-I (समीकरण)** | **Column-II (व्यापक हल)** |
|---|---|
| (A) $\frac{1-\cos 2x}{\sin 2x}=0$ | (P) $(8n-1)\frac{\pi}{4}, n \in I$ |
| (B) $\cos^2 x + 5\cos x = 2\sin^2 x$ | (Q) $2n\pi \pm \cos^{-1}\frac{1}{3}, n \in I$ |
| (C) $\sin^4 x + \sin^4\left(\frac{x}{2}+\frac{\pi}{8}\right) + \cos^4 x = \frac{1}{2}\sin^2 2x$ | (R) कोई हल नहीं |
| (D) $4\cos^2\theta \cdot \sin\theta - 2\sin^2\theta = 3\sin\theta$ | (S) $n\pi + (-1)^n\left(-\frac{3\pi}{10}\right), n \in I$ |

**Ans.** (A) R, (B) Q, (C) P (D) S

**Sol.** (A) $\frac{1-\cos 2x}{\sin 2x}=0$

$\Rightarrow \frac{2\sin^2 x}{2\sin x\cos x}=0 \qquad \{\because \sin 2x \neq 0\}$

$\Rightarrow \tan x = 0$

$\Rightarrow x = n\pi; n \in I$

$\because \sin 2x \neq 0$

$\Rightarrow$ This equation has no solution.

(B) $\cos^2 x + 5\cos x = 2\sin^2 x$

$\Rightarrow \cos^2 x + 5\cos x - 2(1-\cos^2 x) = 0$

$\Rightarrow 3\cos^2 x + 5\cos x - 2 = 0$

$\Rightarrow \cos x = -2$ (Not possible as $\cos x \in [-1, 1]$) or $\cos x = \frac{1}{3}$

$\Rightarrow x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right); n \in I$

(C) $\sin^4 x + \cos^4 x + \sin^4\left(\frac{x}{2}+\frac{\pi}{8}\right) = \frac{1}{2}\sin^2 2x$

$\Rightarrow 1 - \sin^2 2x + \sin^4\left(\frac{x}{2}+\frac{\pi}{8}\right) = 0$

$\Rightarrow \cos^2 2x + \sin^4\left(\frac{x}{2}+\frac{\pi}{8}\right) = 0$

$\Rightarrow \cos 2x = 0$ or $\sin\left(\frac{x}{2}+\frac{\pi}{8}\right) = 0$

$\Rightarrow x = n\pi \pm \frac{\pi}{4}$ or $x = 2n\pi - \frac{\pi}{4}; n \in I$

(D) $4\cos^2\theta \sin\theta - 2\sin^2\theta = 3\sin\theta$

$\Rightarrow 4\sin\theta(1 - \sin^2\theta) - 2\sin^2\theta - 3\sin\theta = 0$

$\Rightarrow \sin\theta - 4\sin^3\theta - 2\sin^2\theta = 0$

$\Rightarrow 4\sin^3\theta + 2\sin^2\theta - \sin\theta = 0$

$\Rightarrow \sin\theta(4\sin^2\theta + 2\sin\theta - 1) = 0$

$\Rightarrow \sin\theta = 0$ or $\sin\theta = \frac{\sqrt{5}-1}{4}$ or $\sin\theta = -\left(\frac{1+\sqrt{5}}{4}\right)$

$\Rightarrow \theta = n\pi$ or $\theta = n\pi + (-1)^n \frac{\pi}{10}$ or $\theta = n\pi + (-1)^n\left(-\frac{3\pi}{10}\right)$

23. Match the following column :

| **Column-I** | **Column-II** |
|---|---|
| (A) $\cos 3x \cdot \cos^3 x + \sin 3x \cdot \sin^3 x = 0$ | (P) $n\pi \pm \frac{\pi}{3}$ |
| (B) $\sin 3\alpha = 4\sin\alpha \sin(x+\alpha)\sin(x-\alpha)$ where $\alpha$ is a constant $\neq n\pi$ | (Q) $n\pi + \frac{\pi}{4}, n \in I$ |
| (C) $\lvert 2\tan x - 1\rvert + \lvert 2\cot x - 1\rvert = 2$ | (R) $\frac{n\pi}{4} + \frac{\pi}{8}, n \in I$ |
| (D) $\sin^{10}x + \cos^{10}x = \frac{29}{16}\cos^4 2x$ | (S) $\frac{n\pi}{2} \pm \frac{\pi}{4}$ |

निम्न कॉलम को सुमेलित कीजिए :

| **Column-I** | **Column-II** |
|---|---|
| (A) $\cos 3x \cdot \cos^3 x + \sin 3x \cdot \sin^3 x = 0$ | (P) $n\pi \pm \frac{\pi}{3}$ |
| (B) $\sin 3\alpha = 4\sin\alpha \sin(x+\alpha)\sin(x-\alpha)$ जहाँ $\alpha$ एक नियतांक है $\neq n\pi$ | (Q) $n\pi + \frac{\pi}{4}, n \in I$ |
| (C) $\lvert 2\tan x - 1\rvert + \lvert 2\cot x - 1\rvert = 2$ | (R) $\frac{n\pi}{4} + \frac{\pi}{8}, n \in I$ |
| (D) $\sin^{10}x + \cos^{10}x = \frac{29}{16}\cos^4 2x$ | (S) $\frac{n\pi}{2} \pm \frac{\pi}{4}$ |

**Ans.** (A) S; (B) P; (C) Q; (D) R

**Sol.** (A) $\cos 3x \cos^3 x + \sin 3x \sin^3 x = 0$

$\Rightarrow (4\cos^3 x - 3\cos x)\cos^3 x + (3\sin x - 4\sin^3 x)\sin^3 x = 0$

$\Rightarrow 4(\cos^6 x - \sin^6 x) + 3(\sin^4 x - \cos^4 x) = 0$

$\Rightarrow \cos 2x [4(1 - \sin^2 x \cos^2 x) - 3] = 0$

$\Rightarrow \cos^3 2x = 0$

$\Rightarrow 2x = 2n\pi \pm \frac{\pi}{2}$

$\Rightarrow x = n\pi \pm \frac{\pi}{4}; n \in I$

(B) $\sin 3\alpha = 4 \sin \alpha \sin (x + \alpha) \sin (x - \alpha); a \neq n\pi$

$\because \sin(\alpha) \sin(60^o + \alpha) \sin(60^o - \alpha) = \frac{1}{4}\sin 3\alpha$

$\therefore x = n\pi \pm \frac{\pi}{3}$

(C) $|2 \tan x - 1) + |2 \cot x - 1| = 2$

$\Rightarrow (2 \tan x - 1)^2 + (2 \cot x - 1)^2 + 2 |(2 \tan x - 1)(2 \cot x - 1)| = 4$

$\Rightarrow 4(\tan^2 x + \cot^2 x) + 2 - 4 \tan x - 4 \cot x + 2 |4 \tan x \cot x - 2(\tan x + \cot x) + 1| = 4$

Let $\tan x + \cot x = t$

$\Rightarrow 4 (t^2 - 2) + 2 - 4t + 2 |5 - 2t| = 4$

$4t^2 - 8 + 2 - 4t + 2|5 - 2t| = 4$

Case-I : If $5 - 2t > 0$ i.e. $t < \frac{5}{2}$

Then

$4t^2 - 8 + 2 - 4t + 10 - 4t - 4 = 0$

$4t^2 - 8t = 0$

$\Rightarrow \underbrace{t = 0}_{\text{Not possible}}$ or $t = 2$

$\Rightarrow x = n\pi + \frac{\pi}{4}; n \in I$

Case-II : If $5 - 2t < 0$ i.e. $t > \frac{5}{2}$

then

$4t^2 - 8 + 2 - 4t - 10 + 4t - 4 = 0$

$\Rightarrow 4t^2 = 20$

$\Rightarrow t^2 = 5$

$\because$ not possible as $t > \frac{5}{2}$

(D) $\sin^{10} x + \cos^{10} x = \frac{29}{16}\cos^4 2x$

$$\Rightarrow \left(\frac{1-\cos 2x}{2}\right)^5 + \left(\frac{1+\cos 2x}{2}\right)^5 = \frac{29}{16}(\cos 2x)^4$$

let $\cos 2x = t$

$$\Rightarrow \left(\frac{1-t}{2}\right)^5 + \left(\frac{1+t}{2}\right)^5 = \frac{29}{16}t^4$$

$\Rightarrow (1 - t)^5 + (1 + t)^5 = 58t^4$

$\Rightarrow 2 + 20t^2 + 10t^4 - 58t^4 = 0$

$\Rightarrow 1 + 10t^2 - 24t^4 = 0$

$\Rightarrow 24t^4 - 10t^2 - 1 = 0$

$\Rightarrow (2t^2 - 1)(12t^2 + 1) = 0$

$\Rightarrow 2t^2 - 1 = 0$ or $12t^2 + 1 = 0$ (not possible)

$$\Rightarrow \cos^2 2x = \frac{1}{2}$$

$\Rightarrow \cos 4x = 0$

$$\Rightarrow 4x = n\pi + \frac{\pi}{2}$$

$$\Rightarrow x = \frac{n\pi}{4} + \frac{\pi}{8}; n \in I$$